

Principal Component Regression

A Biased Regression Technique

Multicollinearity

- Two or more explanatory variables in a regression model are highly correlated
- Causes inflated standard errors on the regression coefficients
- Can potentially switch the expected sign of a regression coefficient

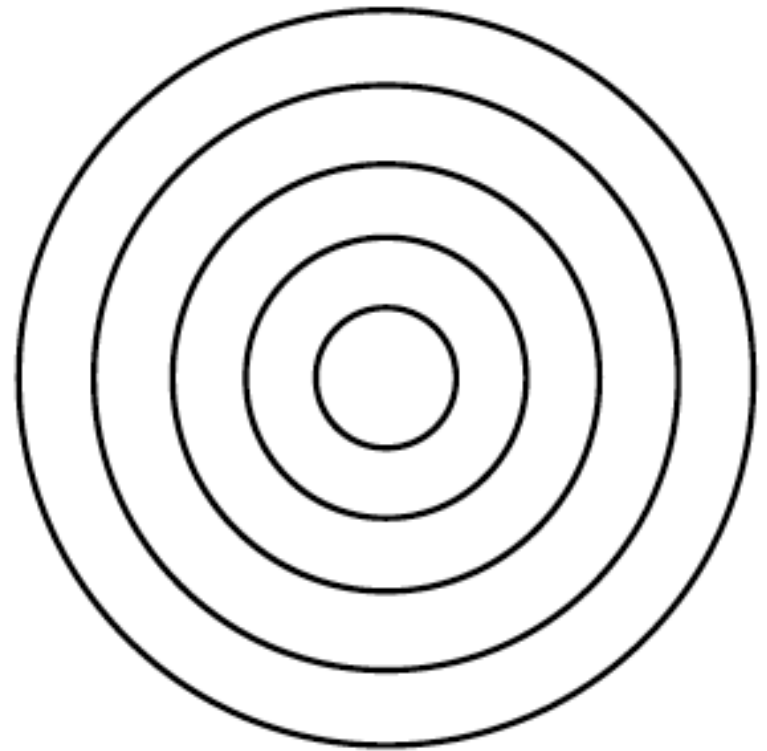
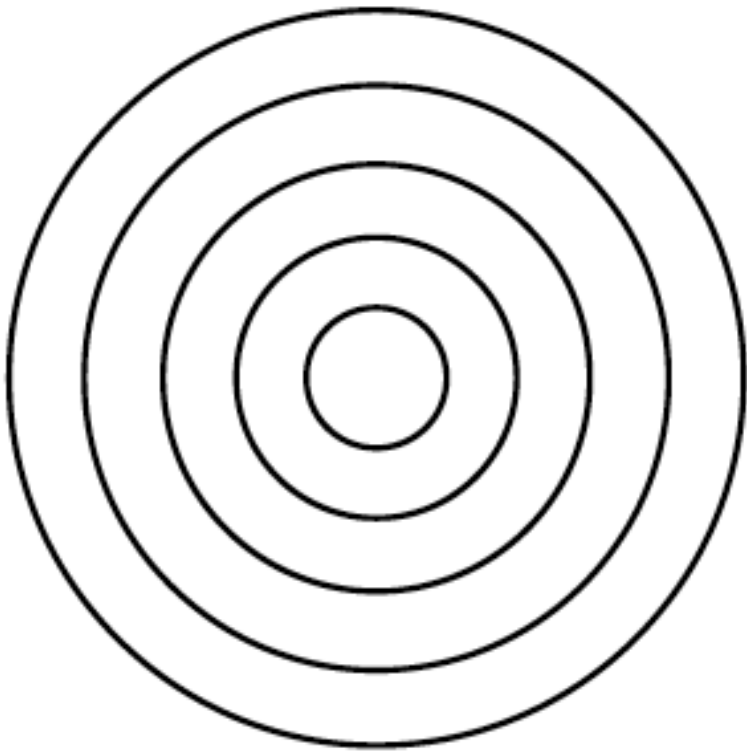
Dealing w/ Multicollinearity

- Exclude redundant independent variables
- Redefine variables
- For polynomial regression, center the independent variables
- **Use biased regression techniques**
 - Principal Component Regression
 - Ridge Regression (Later)

Biased Regression Techniques

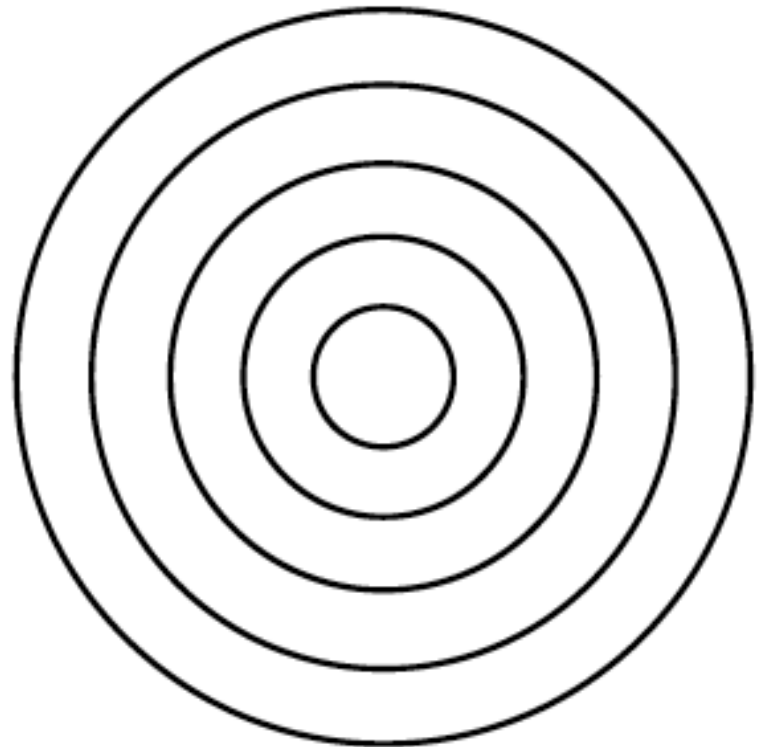
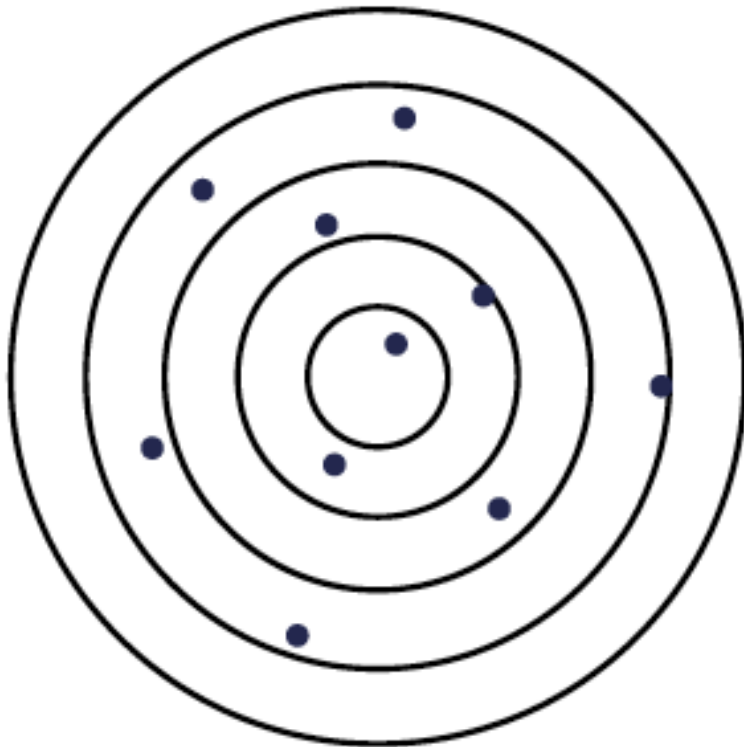
- Biased regression techniques intentionally bias the estimation of the regression coefficients.
- Although these methods produce estimates that are biased, they tend to have more precision than OLS estimates ***in the presence of multicollinearity***.

Biased Regression Techniques



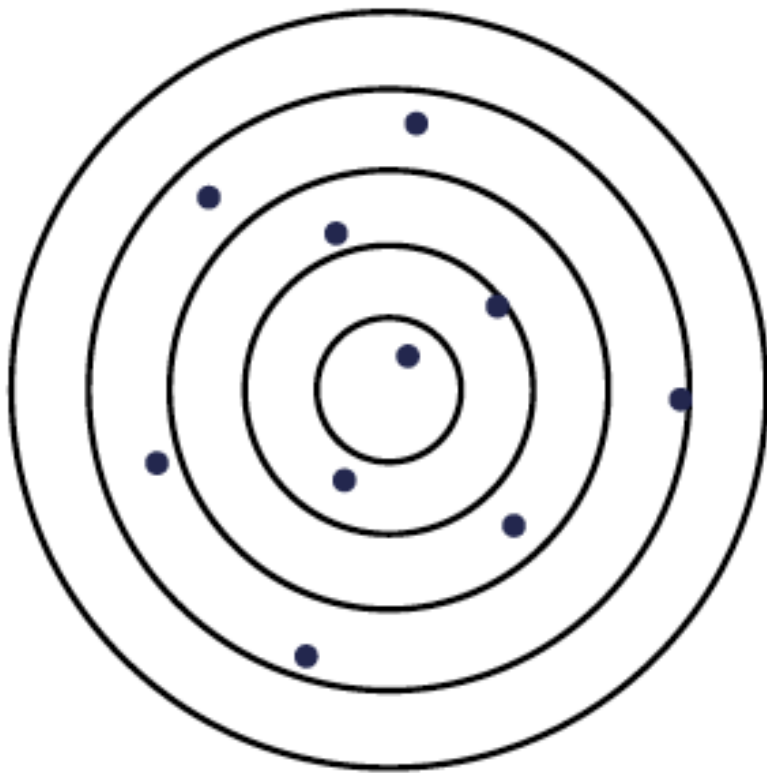
Biased Regression Techniques

Unbiased but not precise

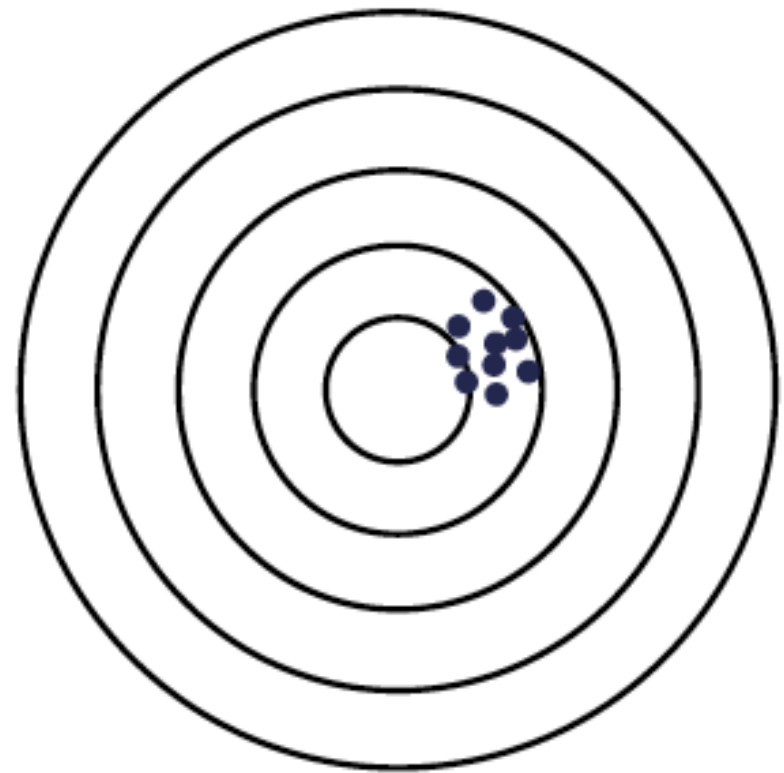


Biased Regression Techniques

Unbiased but not precise

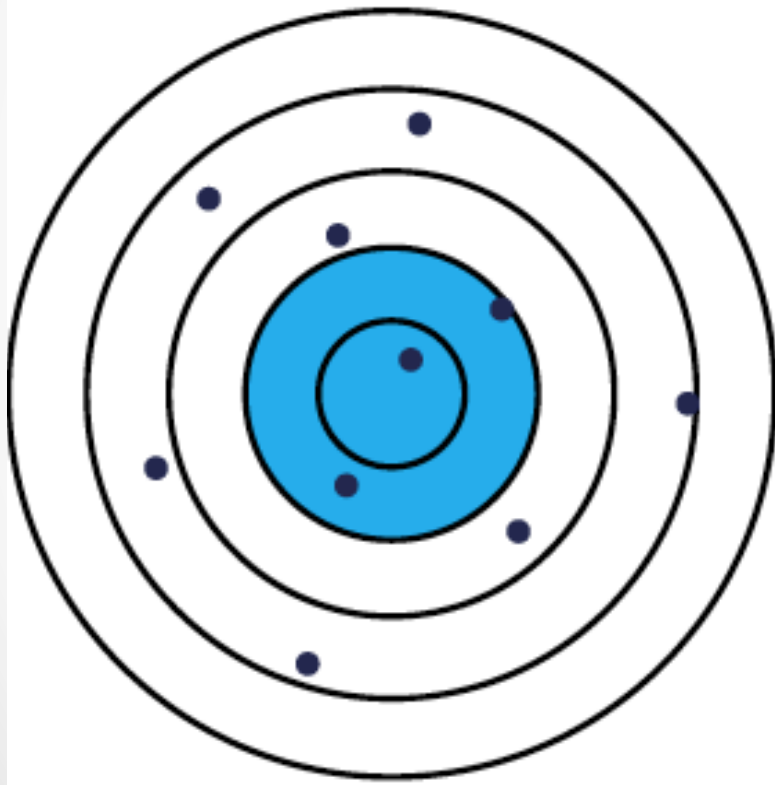


Biased but precise

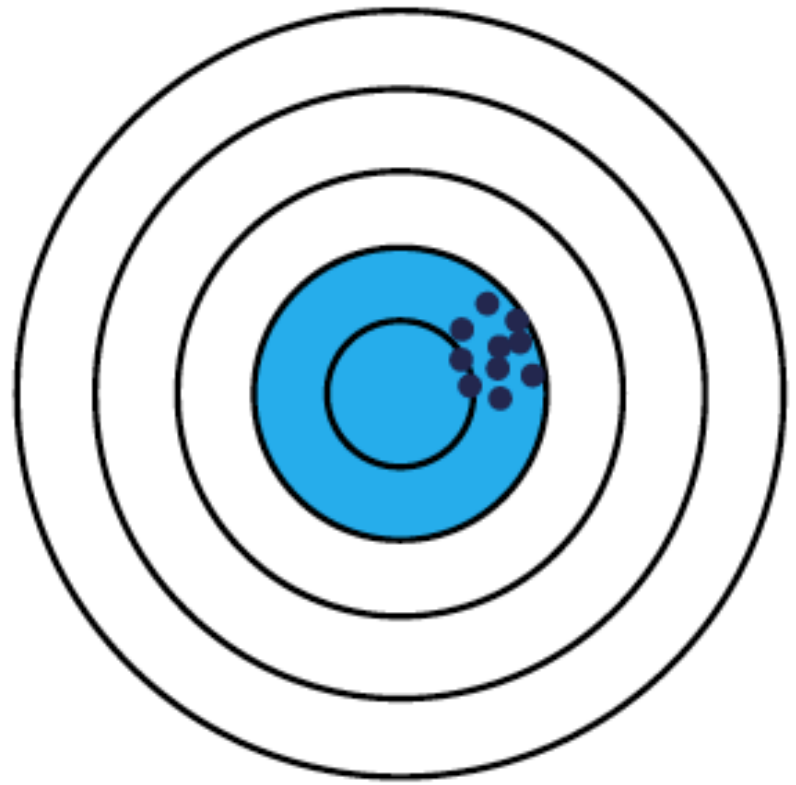


Biased Regression Techniques

Unbiased but not precise



Biased but precise



Principal Component Regression

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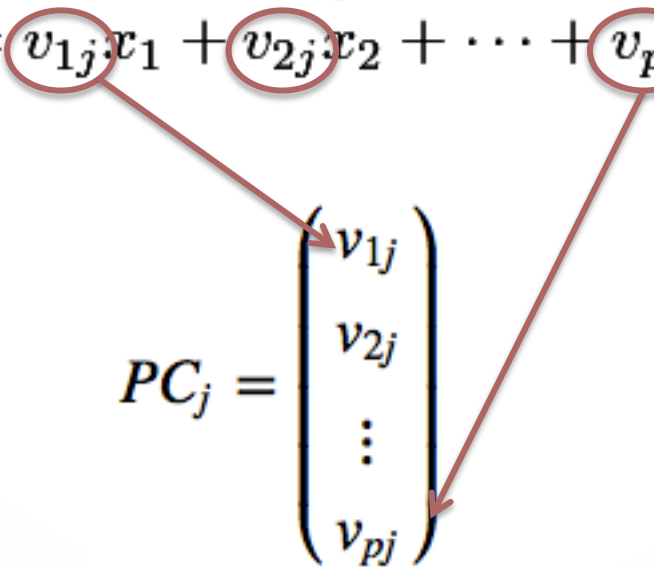
Principal Components

- Multicollinearity is a problem where variables are highly correlated.
- Any collection of variables can be recreated as set of orthogonal predictor variables that are a linear combination of the original variables.
- The orthogonal predictor variables are the **principal components**.

Principal Components

- Recall that principal components are simply linear combinations of original variables.
- The coefficients of that linear combination are typically reflected in a vector:

$$PC_j = v_{1j}x_1 + v_{2j}x_2 + \cdots + v_{pj}x_p$$

$$PC_j = \begin{pmatrix} v_{1j} \\ v_{2j} \\ \vdots \\ v_{pj} \end{pmatrix}$$
The diagram illustrates the relationship between the coefficients in the linear combination and a vector. Two red arrows originate from the circled terms $v_{1j}x_1$ and $v_{pj}x_p$ in the equation above. The arrow from $v_{1j}x_1$ points to the top element v_{1j} of the vector, and the arrow from $v_{pj}x_p$ points to the bottom element v_{pj} of the vector. This visualizes how the coefficients are organized into a single column vector.

Principal Components

- Covariance Matrix of PCs looks like the following:

$$\begin{array}{l} PC_1 \\ PC_2 \\ \vdots \\ PC_p \end{array} \longrightarrow \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda_p \end{pmatrix}$$

Principal Components

- Covariance Matrix of PCs looks like the following:

$$\begin{array}{l} PC_1 \longrightarrow \\ PC_2 \longrightarrow \\ \vdots \\ PC_p \longrightarrow \end{array} \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \lambda_p \end{pmatrix}$$

Variances of the PC's

- If the value of $\lambda_i = 0$, then the corresponding PC has no variance (a.k.a. a constant).

Principal Components Regression

- By omitting any PC with λ_j close to 0, we essentially create an approximation of our data which has no multicollinearity in the regression model.
- WHAT IS THE PROBLEM IF WE DO NOT OMIT ANY PRINCIPAL COMPONENTS FROM OUR ANALYSIS?

Example – French Economy

- We are going to examine data from the French economy reported by Malinvaud (1968).
- Variables:
 - Imports
 - Domestic Production
 - Stock Formation
 - Domestic Consumption
- All measured in billions of French francs between 1949 and 1966.

Example – French Economy

```
proc reg data=Advanced.French;  
    model Import = DoProd Stock Consum /vif;  
run;  
quit;  
  
proc princomp data=Advanced.French out=frenchPC;  
    var DoProd Stock Consum;  
run;
```


Example – French Economy

Eigenvalues of the Correlation Matrix

	Eigenvalue	Difference	Proportion	Cumulative
1	1.99915493	1.00100076	0.6664	0.6664
2	0.99815418	0.99546329	0.3327	0.9991
3	0.00269089		0.0009	1.0000

Eigenvectors

	Prin1	Prin2	Prin3
DoProd	0.706330	-.035689	0.706982
Stock	0.043501	0.999029	0.006971
Consum	0.706544	-.025830	-.707197

Example – French Economy

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Example – French Economy

- Now there is a “new” set of variables (the PC’s) that are orthogonal to each other, which eliminates multicollinearity concerns.
- As long as you remove those that are nearly constant!
- Principal Components Regression (PCR) Model:

$$\tilde{Y} = \alpha_1 PC_1 + \alpha_2 PC_2 + \cdots + \alpha_k PC_k + \epsilon$$

Example – French Economy

```
proc princomp data=Advanced.French out=frenchPC;  
    var DoProd Stock Consum;  
run;  
  
proc standard data=frenchPC mean=0 std=1 out=frenchPC2;  
    var Import;  
run;  
  
proc reg data=frenchPC2;  
    Reg: model Import = DoProd Stock Consum /vif;  
    PCa: model Import = Prin1 Prin2 Prin3 /vif;  
    PCb: model Import = Prin1 Prin2 /vif;  
run;  
quit;
```

Example – French Economy

```
proc princomp data=Advanced.French out=frenchPC;  
    var DoProd Stock Consum;  
run;  
  
proc standard data=frenchPC mean=0 std=1 out=frenchPC2;  
    var Import;  
run;  
  
proc reg data=frenchPC2;  
    Reg: model Import = DoProd Stock Consum /vif;  
    PCa: model Import = Prin1 Prin2 Prin3 /vif;  
    PCb: model Import = Prin1 Prin2 /vif;  
run;  
quit;
```

```
graph TD; A["out=frenchPC"] --> B["data=frenchPC"]; C["out=frenchPC2"] --> D["data=frenchPC2"];
```

Example – French Economy

- PCR model:

$$\tilde{Y} = \alpha_1 PC_1 + \alpha_2 PC_2 + \dots + \alpha_k PC_k + \epsilon$$

- What is the problem with the new variables and coefficients $\alpha_1, \dots, \alpha_k$?

INTERPRETATION!

Principal Components Regression

- Since the PC's are simply functions of the original variables, they can be replaced with the linear combinations.

$$\tilde{Y} = \alpha_1 PC_1 + \alpha_2 PC_2 + \cdots + \alpha_k PC_k + \epsilon$$

$$PC_j = v_{1j}x_1 + v_{2j}x_2 + \cdots + v_{pj}x_p$$



$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$$

The Replacement Math

$$\tilde{y} = \frac{y - \bar{y}}{s_y} \quad \tilde{x} = \frac{x - \bar{x}}{s_x}$$

$$\tilde{y} = \alpha_1 PC_1 + \alpha_2 PC_2 + \cdots + \alpha_p PC_p + \epsilon$$

$$PC_j = \mathbf{V}_{1j}\tilde{x}_1 + \mathbf{V}_{2j}\tilde{x}_2 + \cdots + \mathbf{V}_{pj}\tilde{x}_p$$

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$$

$$\beta_j = \frac{s_y}{s_{x_j}} (\mathbf{V}_{j1}\alpha_1 + \mathbf{V}_{j2}\alpha_2 + \cdots + \mathbf{V}_{jp}\alpha_p)$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}_1 - \beta_2 \bar{x}_2 - \cdots - \beta_p \bar{x}_p$$

Automatic Replacement via PROC PLS

```
proc pls data=Advanced.French method=pcr nfac=2;  
    model Import = DoProd Stock Consum / solution;  
run;  
quit;
```

Cautions

- May not always work
 - May have trouble explaining variability in the response variable
 - Never blindly drop PCs!
- Outliers / influential observations can severely distort the principal components because they alter the covariance matrix.